Digital Electronics
III Semester
B. Tech. (CSE)
Amrita School of Engineering, Bengaluru
FF conversion and Registers

Flip-Flop Conversion

Steps for Flip-Flop Conversions:

1. Identify available and required flip flop.
2. Write characteristic table for required flip flop.
3. Write excitation table for available flip flop.
4. Make conversion table and using K-map write Boolean expression for available flip flop.
5. Draw the circuit.

Conversion of a JK-to-D Flip-Flop

1. Available Flip Flop : JK Flip Flop
   Required Flip Flop : D Flip Flop

2. Characteristic Table of Required FF: D Flip Flop

Inputs	Present Output	Next Output
D	Q_n	Q_{n+1}
0	0	0
0	1	0
1	0	1
1	1	1

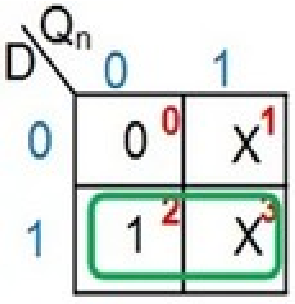
3. Excitation Table of Available FF: JK FF

Q	Q+	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

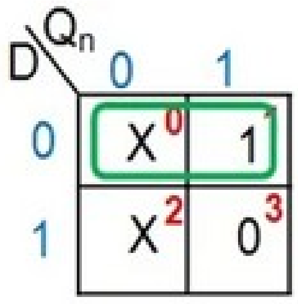
Excitation of J-K flip-flop

4. Conversion Table and Boolean Expression

Qn	D	Qn+1	J	K
0	0	0	0	X
0	1	1	1	X
1	0	0	X	1
1	1	1	X	0

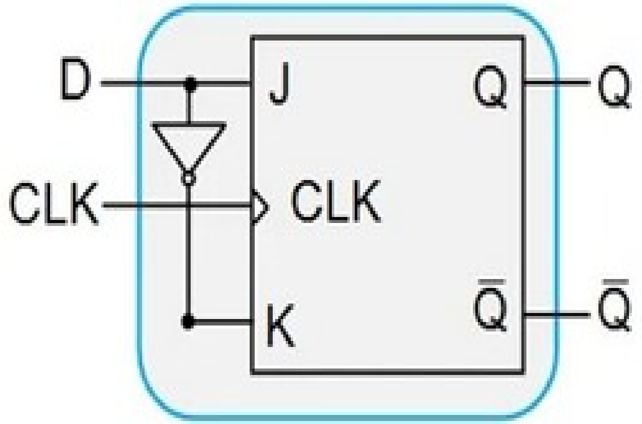


$J = D$



$K = \bar{D}$

5. Circuit Diagram



Truth Table of D Flip-flop

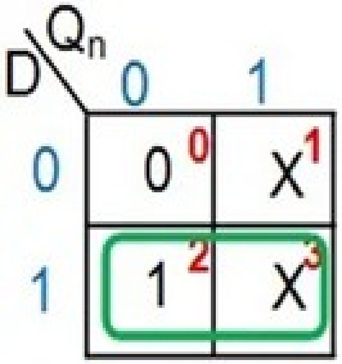
Input	Outputs	
	Present State	Next State
D	Q_n	Q_{n+1}
0	0	0
0	1	0
1	0	1
1	1	1

D Input	Outputs		JK Inputs	
	Present State	Next State		
D	Q_n	Q_{n+1}	J	K
0	0	0	0	X
0	1	0	X	1
1	0	1	1	X
1	1	1	X	0

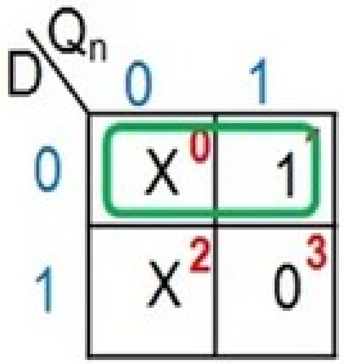
Excitation Table of JK Flip-flop

Outputs		Inputs	
Present State	Next State		
Q_n	Q_{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

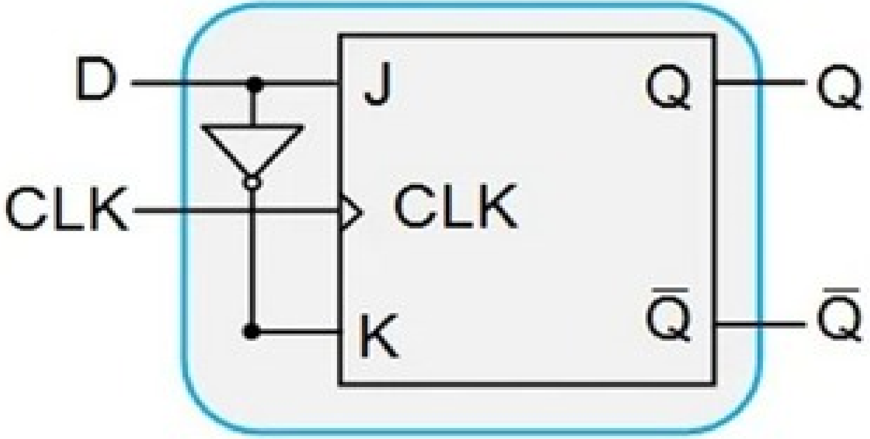
JK to D Conversion Table



$J = D$



$K = \overline{D}$



Conversion of D to JK Flip-Flop

Truth Table of JK Flip-flop

Inputs		Outputs	
		Present State	Next State
J	K	Q_n	Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

JK Inputs		Outputs		D Input
		Present State	Next State	
J	K	Q_n	Q_{n+1}	D
0	0	0	0	0
0	0	1	1	1
0	1	0	0	0
0	1	1	0	0
1	0	0	1	1
1	0	1	1	1
1	1	0	1	1
1	1	1	0	0

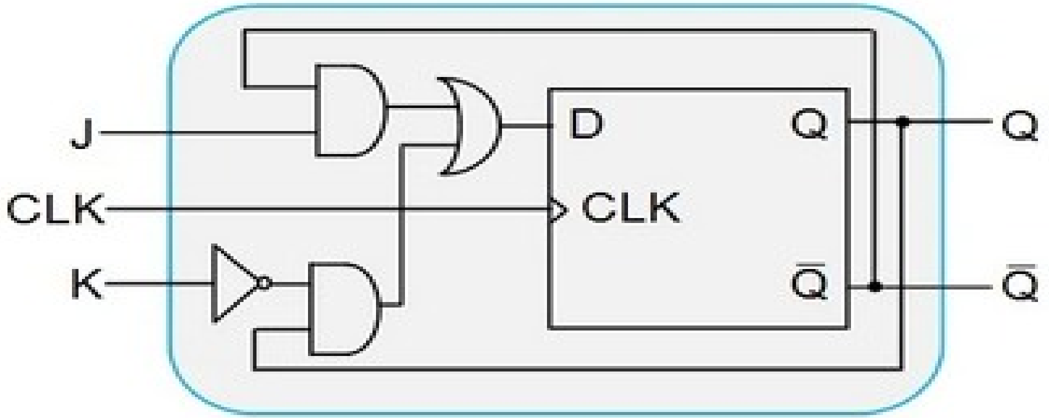
D to JK Conversion Table

Excitation Table of D Flip-flop

Outputs		Input
Present State	Next State	
Q_n	Q_{n+1}	D
0	0	0
0	1	1
1	0	0
1	1	1

J \ KQ_n	00	01	11	10
0	0 [0]	1 [1]	0 [3]	0 [2]
1	1 [4]	1 [5]	0 [7]	1 [6]

$$D = J\bar{Q}_n + \bar{K}Q_n$$



Try Yourself:

1. SR FF to JK FF
2. D FF to T FF
3. T FF to D FF
4. JK FF to T FF

Convert a positive edge triggered JK flip-flop to a positive edge triggered XY flip-flop as defined by function table:

X	Y	Qn+1
0	0	0
0	1	Q’
1	0	Q
1	1	1

The application table for the JK flip-flop and function table for the XY f1ip-flop are:

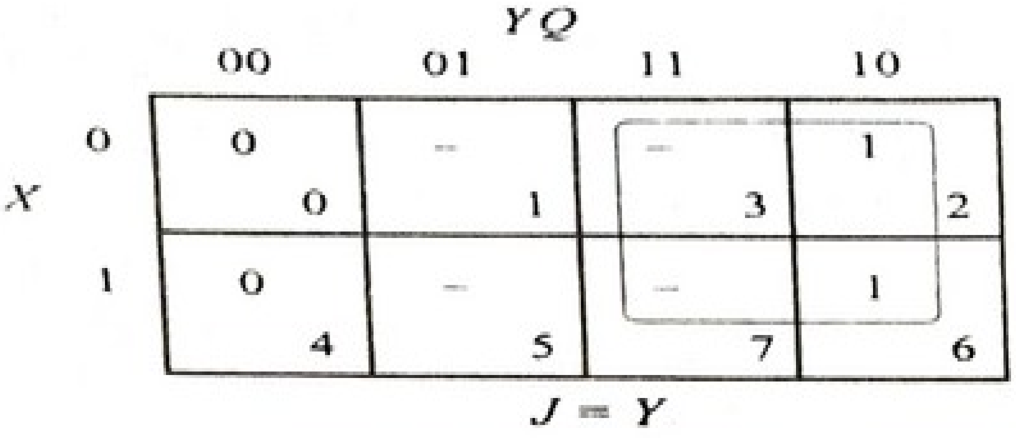
Q →	Qn+1	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

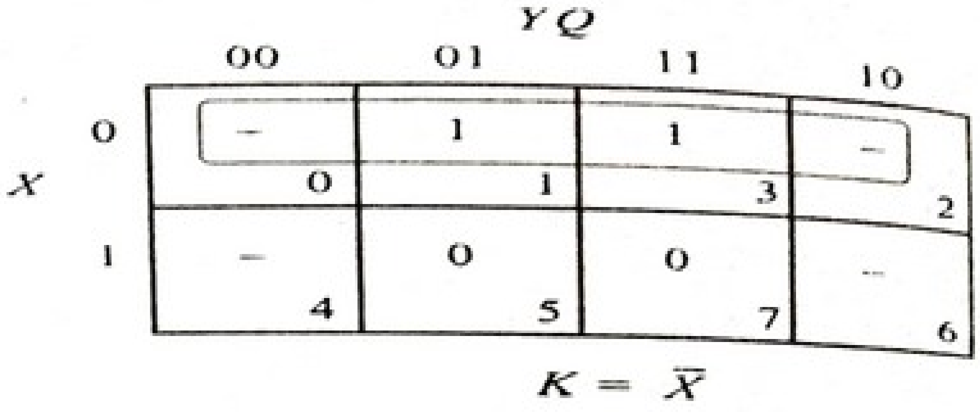
X	Y	Qn+1
0	0	0
0	1	Q’
1	0	Q
1	1	1

Excitation Table:

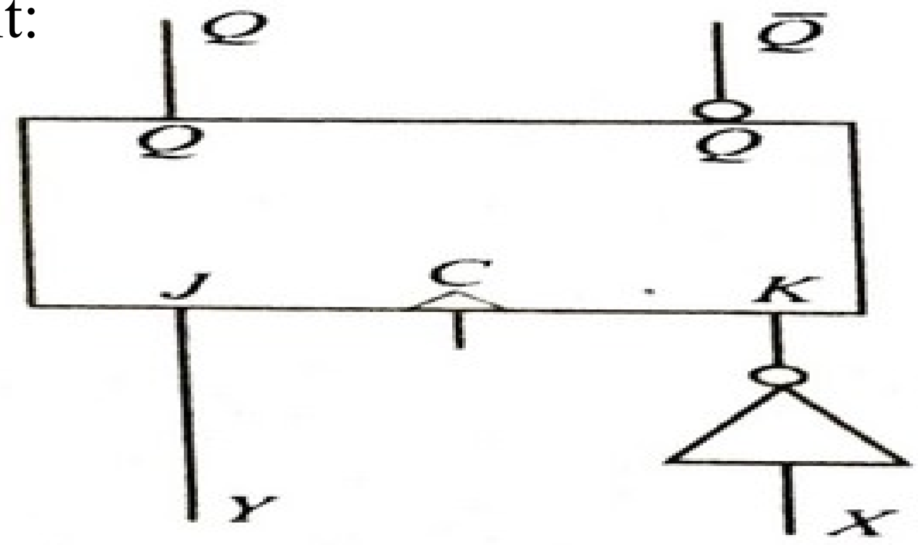
Cell No.	X	Y	Q	Qn+1	J	K
0	0	0	0	0	0	X
1	0	0	1	0	X	1
2	0	1	0	1	1	X
3	0	1	1	0	X	1
4	1	0	0	0	0	X
5	1	0	1	1	X	0
6	1	1	0	1	1	X
7	1	1	1	1	X	0

Karnaugh Map for the J and K inputs





Converted Circuit:



Convert a positive edge triggered D flip-flop into the positive edge triggered XY flip-flop defined by below function table:

X	Y	Qn+1
0	0	0
0	1	Q’
1	0	Q
1	1	1

Solution: The application table for the D flip-flop and the function table for the XY flip-flop are:

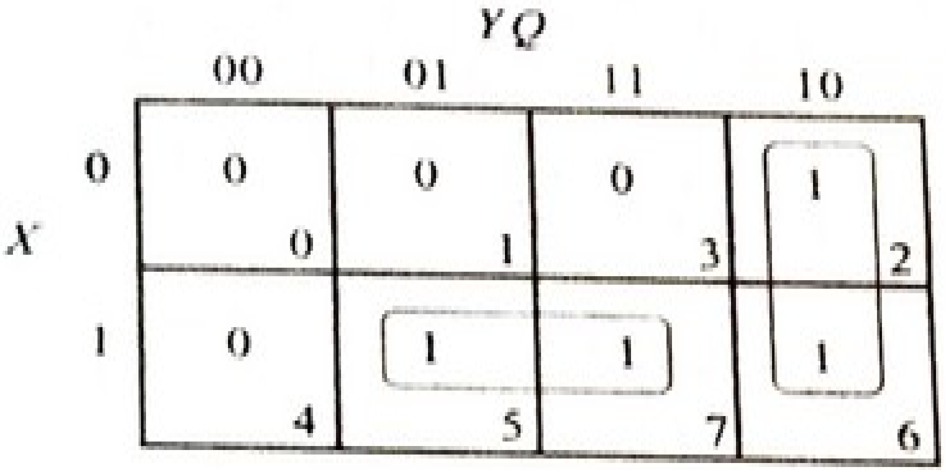
Q $\rightarrow$	Qn+1	D
0	0	0
0	1	1
1	0	0
1	1	1

X	Y	Qn+1
0	0	0
0	1	Q’
1	0	Q
1	1	1

Excitation Table:

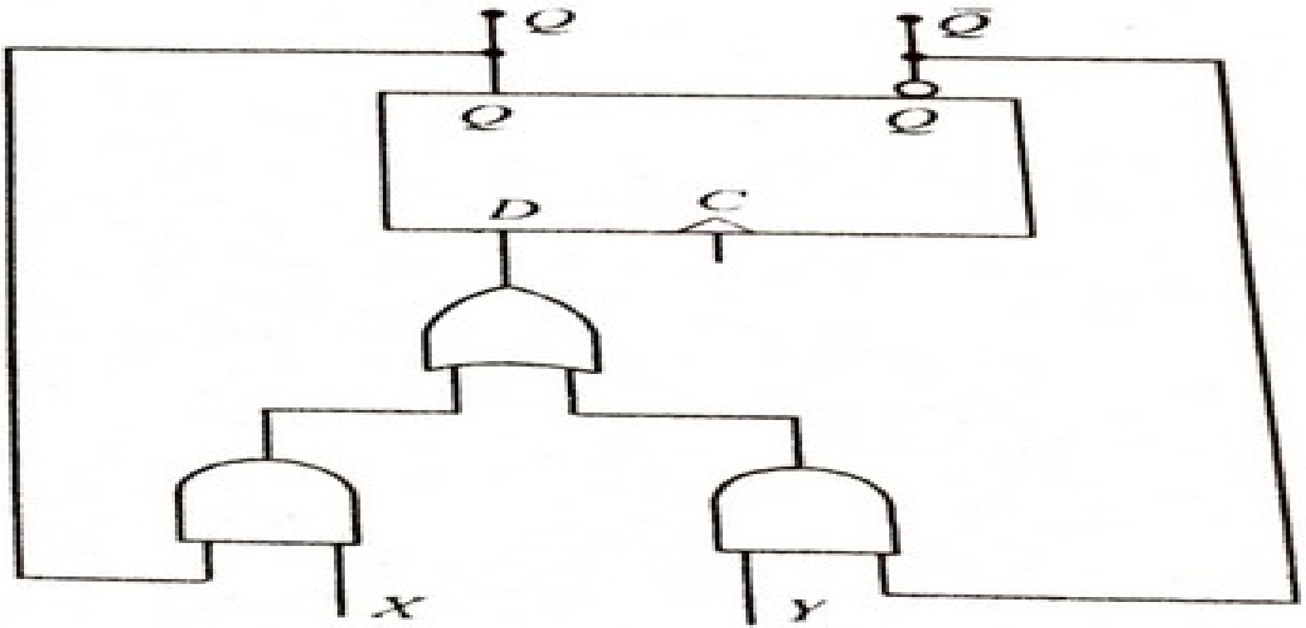
Cell no.	X	Y	Q	→ Q'	D
0	0	0	0	0	0
1	0	0	1	0	0
2	0	1	0	1	1
3	0	1	1	0	0
4	1	0	0	0	0
5	1	0	1	1	1
6	1	1	0	1	1
7	1	1	1	1	1

Karnaugh Map for D input:



$$D = XQ + Y\overline{Q}$$

Converted Circuit:



A PN flip-flop has four operations: clear to 0, no change, complement, and set to 1, when inputs P and N are 00, 01, 10, and 11, respectively.

a)Tabulate the characteristic table.

b)Derive the characteristic equation.

c)Tabulate the excitation table.

d)Show how the PN flip-flop can be converted to a D flip-flop.

P	N	Q(t+1)
0	0	0
0	1	Q(t)
1	0	Q′(t)
1	1	1

P	N	Q(t)	Q(t+1)
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1

N

P

P\NQ	00	01	11	10
0	0	0	1	0
1	1	0	1	1

$$Q(t+1) = PQ' + NQ$$

Excitation Table:

Q(t)	Q(t+1)	P	N
0	0	0	X
0	1	1	X
1	0	X	0
1	1	X	1

PN flip-flop can be converted to a D flip-flop, By connecting P and N together.

$$Q(t+1) = DQ' + DQ = D$$

REGISTERS

A register is a group of flip-flops capable of storing one bit of information.

Register consists of a group of flip-flops & gates that effect their transition. The flip-flops holds the binary information & the gate control when & how new information is transferred into the register.

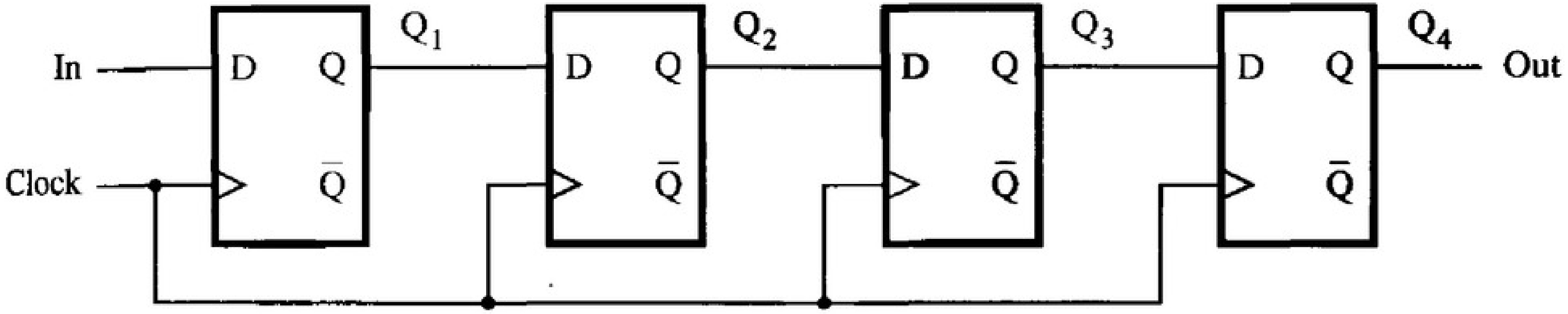
The simplest register is that which only contains flip-flops , with no external gates.

SHIFT REGISTERS

- In digital circuits a **shift register** is a group of flip flops set up in a linear fashion which have their inputs and outputs connected together in such a way that the data are shifted down the line when the circuit is activated.

- The simplest shift register is one that uses only flip-flops.

A Simple Shift Register



- Figure shows a four-bit shift register that is used to shift its contents one bit-position to the right.
- The data bits are loaded into the shift register in a serial fashion using the In input.
- The contents of each flip-flop are transferred to the next flip-flop at each positive edge of the clock.

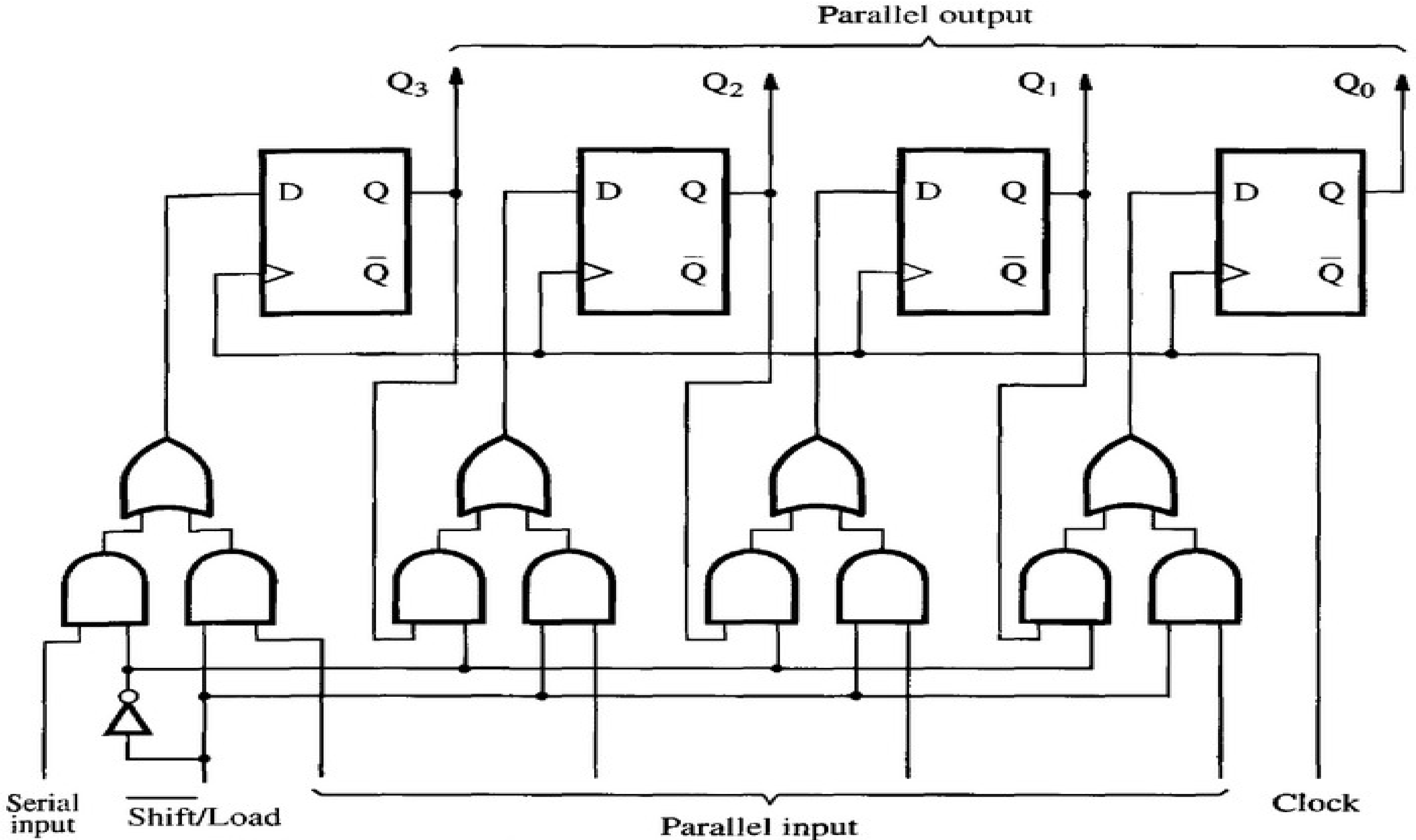
- An illustration of the transfer is shown below, which shows what happens when the signal values at In during eight consecutive clock cycles are 1,0,1,1,1,0,0,and 0, assuming that the initial state of all flip-flops is 0.

	In	Q_1	Q_2	Q_3	Q_4 = Out
t_0	1	0	0	0	0
t_1	0	1	0	0	0
t_2	1	0	1	0	0
t_3	1	1	0	1	0
t_4	1	1	1	0	1
t_5	0	1	1	1	0
t_6	0	0	1	1	1
t_7	0	0	0	1	1

- To implement a shift register, it is necessary to use either edge-triggered or master-slave flip-flops. The level-sensitive gated latches are not suitable, because a change in the value of In would propagate through more than one latch during the time when the clock is equal to 1.

PARALLEL-ACCESS SHIFT REGISTER

In computer systems it is often necessary to transfer n-bit data items. This may be done by transmitting all bits at once using n separate wires, in which case we say that the transfer is performed in *parallel*. But it is also possible to transfer all bits using a single wire, by performing the transfer one bit at a time, in n consecutive clock cycles. We refer to this scheme as *serial* transfer. To transfer an n-bit data item serially, we can use a shift register that can be loaded with all n bits in parallel (in one clock cycle). Then during the next n clock cycles, the contents of the register can be shifted out for serial transfer. The reverse operation is also needed. If bits are received serially, then after n clock cycles the contents of the register can be accessed in parallel as an n-bit item.



Parallel output
Q_3
Q_2
Q_1
Q_0
D
Q
$\bar{Q}$
Serial input
$\overline{\text{Shift}}$/Load
Parallel input
Clock

- Figure shows a four-bit shift register that allows the parallel access.

- The control signal $\overline{Shift}/Load$ is used to select the mode of operation. If $\overline{Shift}/Load$ = 0, then the circuit operates as a shift register. If $\overline{Shift}/Load$ = 1, then the parallel input data are loaded into the register.

- In both cases the action takes place on the positive edge of the clock.

- A circuit in which data can be loaded in series and then accessed in parallel is called a series-to-parallel converter. Similarly, the opposite type of circuit is a parallel-to-series converter. The circuit shown can perform both of these functions.